University of Rajshahi

Department of Computer Science and Engineering

CSE4261 Neural Networks and Deep Learning

Assignment-5

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1 Introduction

Deep neural networks, despite their superhuman performance on many tasks, have been shown to be surprisingly vulnerable to adversarial attacks. An adversarial attack involves making small, often imperceptible, perturbations to an input image with the goal of causing the model to misclassify it. This highlights a critical aspect of model robustness and security.

This report demonstrates one of the earliest and most well-known "white-box" attacks: the **Fast Gradient Sign Method (FGSM)**. A white-box attack assumes the attacker has full access to the model's architecture and weights. We will implement FGSM to attack a **MobileNetV2** model pre-trained on the ImageNet dataset.

The core objectives are:

1. To create an adversarial example from a chosen ImageNet-class image that successfully fools the MobileNetV2 model.

2. To visualize the original image, the perturbation, and the final adversarial image.

3. To discuss the fundamental difference between a targeted FGSM attack and adding simple, unstructured noise from a Gaussian distribution.

2 Methodology

2.1 Model and Image Selection

- **Model: MobileNetV2**, pre-trained on the ImageNet dataset, was chosen as the target for the attack.

- **Image:** A sample image of a [**Your Chosen Image Class, e.g., Samoyed**] was selected. This class is one of the 1000 classes in the ImageNet dataset. The image was preprocessed according to MobileNetV2's requirements (e.g., resized to 224x224 and normalized).

2.2 The Fast Gradient Sign Method (FGSM)

FGSM works by exploiting the model's own gradients. The core idea is to find the direction in which to change the input image's pixels to maximize the loss for the correct class. By taking a small step in that direction, we can push the image just across the model's decision boundary, causing a misclassification.

The process is defined by the following equation:

$$\text{adv\_x} = x + \epsilon \cdot \text{sign}(\nabla_x L(\theta, x, y)) \tag{1}$$

Where:

- **adv_x** is the adversarial image.

- **x** is the original input image.

- **y** is the true label of the input image.

- ϵ (epsilon) is a small scalar that controls the magnitude of the perturbation.

- θ represents the model's parameters.

- $L(\theta, x, y)$ is the loss function (e.g., cross-entropy).

- $\nabla_x L$ is the gradient of the loss with respect to the input image $\mathbf{x}$.

- **sign(...)** takes the sign of each element in the gradient tensor (-1 for negative values, 1 for positive values).

The implementation steps are:

1. Feed the original image through the model to get the initial prediction.
2. Use `tf.GradientTape` to watch the input image.
3. Calculate the loss between the model's prediction and the true label.
4. Use the tape to compute the gradient of the loss with respect to the input image.
5. Extract the sign of the gradients.
6. Create the adversarial image using the FGSM formula above.
7. Feed the new adversarial image to the model to see the new prediction.

3 Implementation and Results

The FGSM attack was executed on the MobileNetV2 model.

3.1 Initial Analysis

First, the original image was passed to the model to establish a baseline prediction.

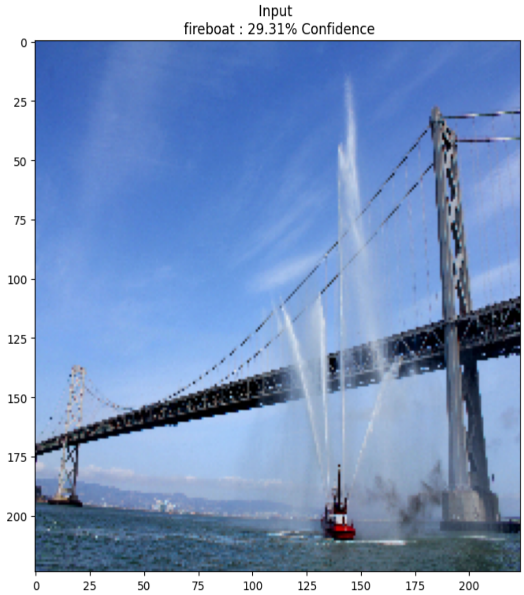
- **Image Class:** The fireboat image.
- **Model's Prediction:** Fireboat
- **Confidence:** 29.31%

The model correctly identified the image with low confidence.

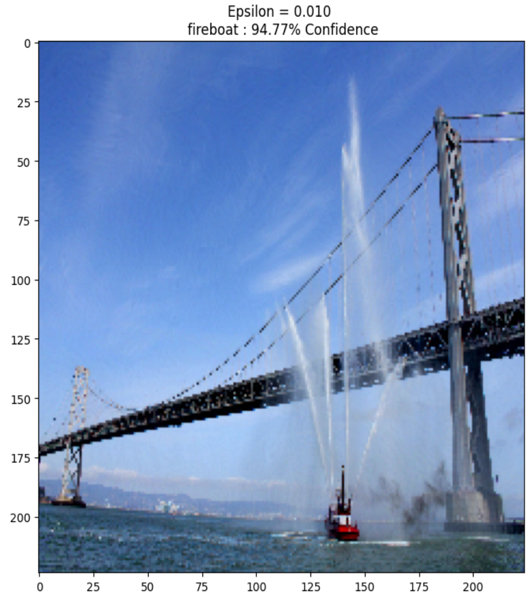
3.2 Adversarial Attack with Epsilon = 0.01

The attack was performed using an epsilon value of **0.01**.

- **New Prediction:** fireboat
- **Confidence:** 94.77%



(a) Original Image



(b) FGSM Perturbation (ϵ=0.01)

3.3 Adversarial Attack with Epsilon = 0.1 and 0.15

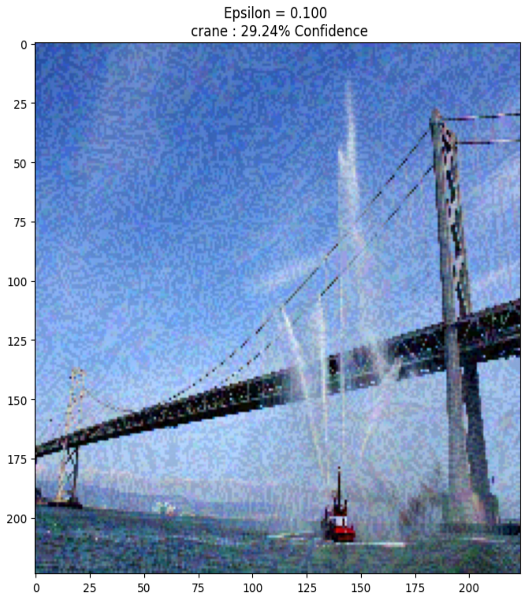
The attack was performed using an epsilon value of **0.1**.

- **New Prediction:** crane
- **Confidence:** 29.24%

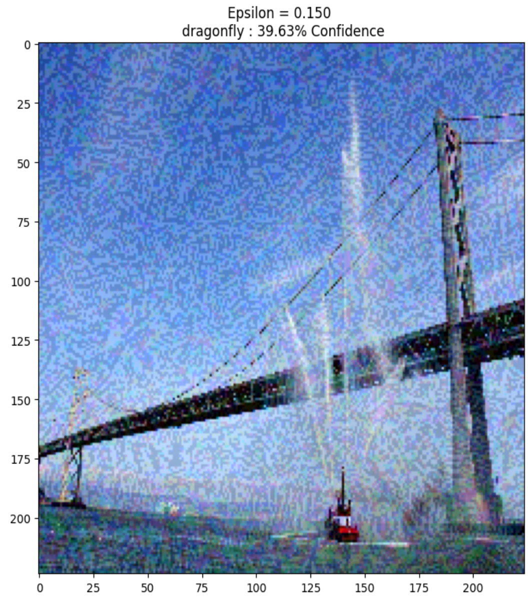
The attack was performed using an epsilon value of **0.15**.

- **New Prediction:** dragonfly
- **Confidence:** 39.63%

As shown in Figure 2, the attack was successful. A visually imperceptible perturbation was enough to cause the model to confidently misclassify the image.



(a) FGSM Perturbation (ϵ=0.1)



(b) FGSM Perturbation (ϵ=0.15)

Figure 2: Visualization of the FGSM attack. The perturbation is scaled for visibility.

4 Discussion: FGSM vs. Gaussian Noise

A key question is whether a similar attack can be performed by simply adding random noise, for instance, from a Gaussian distribution. The answer is **no**, and the reason lies in the fundamental difference between directed and undirected noise.

4.1 FGSM: Directed, Intelligent Noise

FGSM is an *intelligent* attack. The perturbation it creates is not random; it is meticulously crafted to exploit the model's own logic. By calculating the gradient of the loss with respect to the input pixels, we find the exact direction in the high-dimensional image space that will most efficiently increase the loss. This means every modification to every pixel is purposefully pushing the image towards a misclassification. Because this noise is so specifically directed, only a very small amount (controlled by ϵ) is needed to be effective, making the change invisible to the human eye.

4.2 Gaussian Noise: Undirected, Random Noise

Gaussian noise, on the other hand, is completely *undirected* and *random.* The value added to each pixel is drawn from a probability distribution and has no relationship to the model's decision-making process. Adding a small amount of Gaussian noise to an image is highly unlikely to cause a misclassification. The random changes are just as likely to decrease the loss as they are to increase it, and their net effect will likely be negligible.

To fool a model with Gaussian noise, one would have to add a massive amount of it. At that point, the attack is no longer subtle. The image itself would be visibly degraded and noisy. The model's failure would not be due to a clever exploit of its decision boundary, but rather because the input has been distorted beyond recognition. Therefore, adding random noise is not considered an adversarial attack in the same sense as FGSM; it is simply data corruption.

5 Conclusion

This report successfully demonstrated the effectiveness of the Fast Gradient Sign Method in executing a white-box adversarial attack against a state-of-the-art MobileNetV2 model. By adding a small, visually imperceptible, and intelligently crafted perturbation, we were able to force the model to misclassify an image with high confidence.

Furthermore, we established a clear distinction between directed adversarial noise and random noise. An attack like FGSM is powerful because it exploits the model's gradients, making it highly efficient. In contrast, random Gaussian noise is undirected and highly inefficient for attacks, only causing misclassification at levels that visibly corrupt the image. This investigation underscores a critical vulnerability in modern deep learning models and highlights the importance of developing robust defensive mechanisms for real-world AI security.

Assignment Github Link : Assignment